horticulturae

MDPI

Article

WCAF-YOLO: A Lightweight Detection Architecture for Multi-Variety Tomatoes in Unstructured Orchards

Xudong Lin [1], Yihao Zhang [1], Xianzhi Tu [2,*], Zhiguo Du [1], Bin Wen [1], Zhihui Wu [1], Li Yang [1] and Qingwen Wu [2]

[1] College of Mathematics and Informatics, South China Agricultural University, Guangzhou 510642, China; hunanlxd@163.com (X.L.); 20253170162@stu.scau.edu.cn (Y.Z.); bluetooth@scau.edu.cn (Z.D.); wenzip@scau.edu.cn (B.W.); wuzhihui@scau.edu.cn (Z.W.); yl15119085807@outlook.com (L.Y.)

[2] College of Arts, South China Agricultural University, Guangzhou 510642, China; wuqingwen0101@163.com

* Correspondence: xianzhitu@scau.edu.cn

Abstract

Image-level monitoring and variety-level detection of three specialty tomato cultivars, Kiss, Millennium, and White Jade, remain challenging in unstructured orchards because of foliage occlusion, overlapping fruit clusters, and variable illumination. Conventional downsampling may weaken fine spatial details of small targets, whereas larger detectors can impose computational demands that are unsuitable for mobile or edge-based agricultural platforms. To address these limitations, we propose WCAF-YOLO, a lightweight two-dimensional tomato detector based on a modified YOLOv26n architecture. The model replaces the P3 backbone downsampling operation with space-to-depth convolution (SPD-Conv) to retain fine-grained spatial information. Its weighted channel-aware fusion (WCAF) neck combines learnable branch weighting with parameter-free three-dimensional attention to refine fused features. Bounding-box regression uses focaler-minimum point distance intersection over union (Focaler-MPDIoU). Across five random seed runs on the internal held-out test subset of a custom single-site orchard dataset, WCAF-YOLO obtained a mean $mAP_{50–95}$ of 0.9048 ± 0.0013 and a mean recall of 0.9280 ± 0.0019. The corresponding mean improvements over the YOLOv26n baseline were 2.14 and 3.42 percentage points, respectively. The model contained 2.36 M parameters and required 6.36 GFLOPs. Under the evaluated protocol, the model combined a compact parameter count with higher mean detection metrics than the YOLOv26n baseline. The detector outputs two-dimensional bounding boxes and variety labels for image-level orchard monitoring and variety-level assessment. Integration into agricultural field platforms remains to be validated.

Keywords: precision agriculture; tomato detection; orchard monitoring; variety-level detection; lightweight neural networks; YOLOv26n; feature fusion

Check for updates

Academic Editor: Manuela Zude-Sasse

Received: 30 July 2026
Revised: 15 August 2026
Accepted: 18 August 2026
Published: 23 August 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Tomato (*Solanum lycopersicum* L.) is widely cultivated for its nutritional and commercial value. At the study site in Baihe Town, Kaiping City, Guangdong Province, China, Kiss, Millennium, and White Jade tomatoes are commercially important specialty cultivars. Their production still relies on labor-intensive orchard observation, visual harvest assessment, and sorting [1]. These tasks are time-consuming and subject to variation among observers, while repeated manual handling can damage delicate fruit [2]. Lightweight visual perception may therefore assist orchard monitoring, variety-level inspection, and preprocessing for subsequent field assessment under rising labor costs [3,4].

Reliable visual perception remains difficult in unstructured orchards [5]. Dense foliage, overlapping clusters, canopy shadows, and other illumination changes can obscure fruit boundaries [6]. The task also requires discrimination among Kiss, Millennium, and White Jade tomatoes rather than a single fruit–background decision. Within each variety, appearance changes across ripening stages add further variation [7]. These conditions can lead to missed detections, inaccurate boxes, and incorrect variety labels, especially for small or heavily occluded fruit [8,9]. Accordingly, this study develops and evaluates a lightweight two-dimensional detector that localizes individual tomatoes and assigns variety labels in orchard images. The resulting image-level outputs are intended for monitoring and sampling-based field assessment. They are not presented as validated physical yield counts or outputs for robotic grasp planning.

You Only Look Once (YOLO) detectors are widely used for real-time agricultural object detection [10]. Recent variants have addressed tomato detection in complex cultivation environments [11,12]. Related crop-vision studies have examined dense detection, lightweight ripeness assessment, and young-fruit detection under complex backgrounds [13–15]. However, models with large parameter counts and deep feature hierarchies can exceed the thermal and computational budgets of agricultural edge devices [16]. Dense plant canopies also pose a specific representation problem. Strided downsampling reduces spatial resolution, and conventional feature pyramid networks (FPNs) may not explicitly regulate the contributions of features from different scales [17]. Fine boundary cues can therefore become difficult to retain when targets are small, clustered, or partly hidden by foliage.

To address these challenges, we propose WCAF-YOLO, a lightweight detection architecture for unstructured orchard scenes. Built on the YOLOv26n baseline, it modifies selected downsampling, feature-fusion, and regression components while retaining a compact model size. We evaluated the model using 1,998 high-resolution red–green–blue (RGB) images of Kiss, Millennium, and White Jade tomatoes captured with a handheld mobile phone at a commercial orchard in Baihe Town. The dataset was divided into training, validation, and internal held-out test subsets before annotation.

The main contributions of this study are summarized as follows:

- We replace the standard strided convolution at the P3 backbone downsampling stage with space-to-depth convolution (SPD-Conv). This design is intended to reduce the loss of fine-grained spatial information and preserve boundary cues of small and partially occluded tomatoes under dense canopy conditions.
- We design a lightweight weighted channel-aware fusion (WCAF) neck with learnable branch weighting and a parameter-free three-dimensional energy-based attention refinement mechanism. The module is designed to balance multi-scale feature streams and limit background responses from foliage in outdoor orchard scenes.
- We integrate the Focaler-MPDIoU loss function into the bounding box regression process and evaluate its effect on difficult overlapping samples and boundary localization for densely clustered tomato targets.

Accordingly, the objective of this study is to determine whether the combined SPD-Conv, WCAF, and Focaler-MPDIoU design can improve variety-level two-dimensional tomato detection while maintaining a compact computational profile under the evaluated orchard conditions.

2. Materials and Methods

2.1. Overview of the WCAF-YOLO Architecture

To balance variety-level detection accuracy and computational efficiency, we developed WCAF-YOLO, a lightweight detection framework for the complex visual conditions

of tomato canopies. Built upon the YOLOv26n baseline, WCAF-YOLO modifies selected backbone, neck, and loss-function components to address fine-grained variety differentiation under limited computational resources. Figure 1 compares the complete feature-flow topology of the YOLOv26n baseline with that of the proposed model.

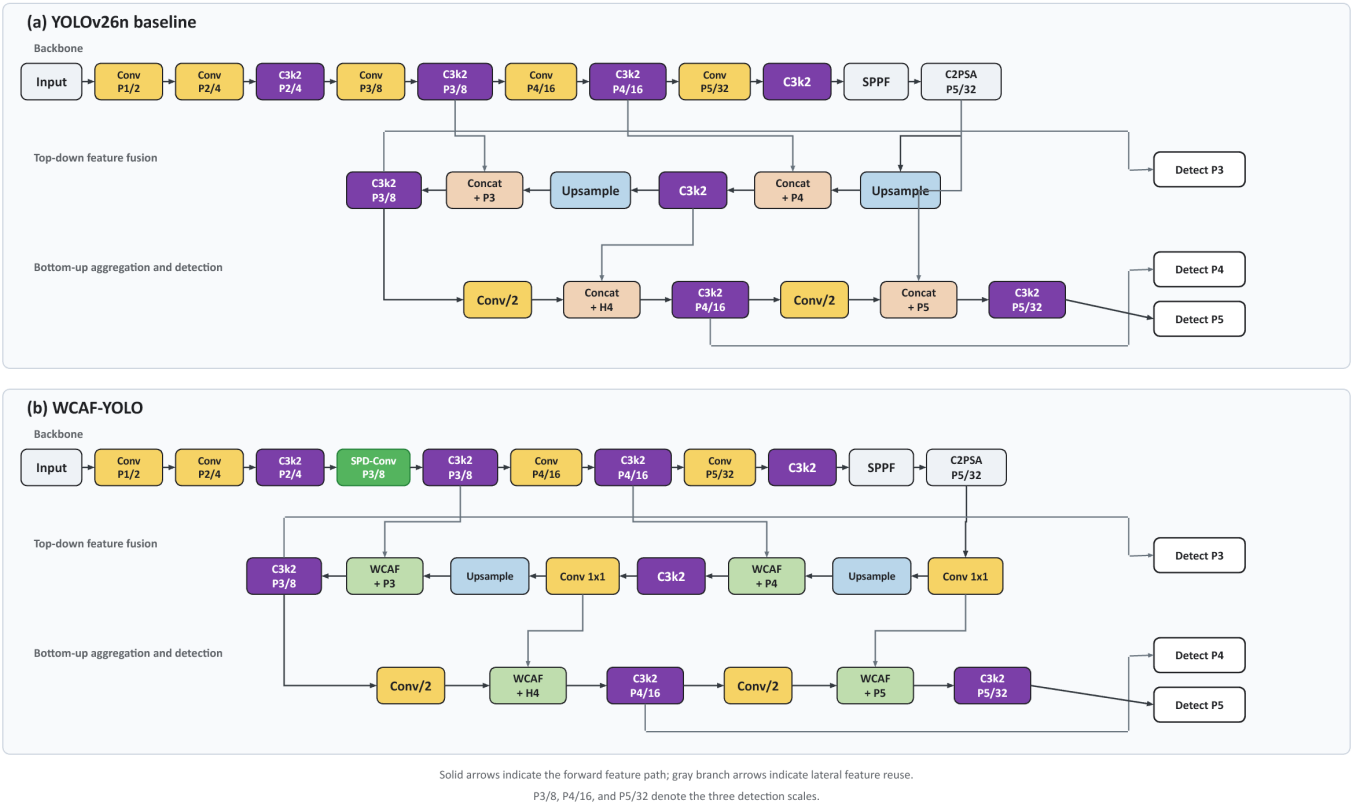


Figure 1. Architectural comparison of (**a**) the YOLOv26n baseline and (**b**) the proposed WCAF-YOLO. Both panels show the backbone, top-down feature fusion, bottom-up aggregation, lateral feature reuse, and the P3/P4/P5 inputs to the detection head. Panel (**b**) identifies the SPD-Conv downsampling operation and the four WCAF fusion nodes introduced in the proposed architecture. Solid arrows denote the forward feature path, and gray branch arrows denote lateral feature reuse. SPPF denotes spatial pyramid pooling–fast; C3k2 and C2PSA retain the module identifiers used in the baseline implementation.

The structural design of WCAF-YOLO centers on three modifications. First, conventional strided convolution at the P3 downsampling stage is replaced with space-to-depth convolution (SPD-Conv) [17]. This operation is intended to retain fine spatial cues before deeper feature extraction. Second, four concatenation nodes in the neck are replaced with weighted channel-aware fusion (WCAF) modules. Each WCAF node combines learnable branch weighting with parameter-free energy-based refinement. Third, focaler-MPDIoU is used for bounding box regression [18]. The resulting detector contains 2.36 M parameters and is evaluated across the three target tomato varieties.

2.2. Dataset Construction and Experimental Setup

2.2.1. Image Acquisition and Annotation

The custom tomato dataset was collected on site to represent unstructured orchard conditions. Images were acquired at a commercial orchard in Baihe Town, Kaiping City, Guangdong Province, China, from October 2025 to March 2026. A handheld mobile phone RGB camera was used rather than a fixed laboratory platform or dedicated robotic system. The phone model was not retained in the acquisition records. The operator moved along

orchard rows and photographed fruit clusters from multiple viewing angles, camera-to-target distances, and canopy positions under natural illumination. The resulting images contain occlusion, overlapping fruit clusters, illumination variation, and background foliage. All images have a native resolution of 3072 × 4096 pixels. The range of viewpoints and distances captures common visual interference and within-variety appearance changes at the collection site.

The images form an orchard scene detection dataset rather than a plant-replicate horticultural experiment. The records therefore do not support reliable reconstruction of the number of plants, repetitions, images per plant, fruit height distribution, or average number of fruits per plant. We report image-level and annotation-level statistics and treat the absence of plant-level replication as a limitation.

Bounding boxes were created with the open-source LabelImg annotation tool (version 1.7). Authors, student annotators, and local fruit growers familiar with the target varieties participated in annotation and review. Each visually separable fruit received an individual box, including fruits in overlapping clusters. Partly occluded fruits and fruits at image boundaries were annotated when the visible region was sufficient to identify both the fruit and its variety. Severely blurred, very small, or otherwise unidentifiable instances were omitted. Visible green immature fruits were assigned to the corresponding target variety rather than to a separate maturity class.

All images underwent a second annotation review, and disagreements were resolved through joint discussion and consensus. The resulting manual boxes serve as the image-level ground truth for the detection experiments. No independent physical count was conducted in the orchard. The evaluation therefore concerns image-level detection rather than validated whole-plant or whole-orchard yield-counting accuracy. Annotations were stored in YOLO format as normalized center coordinates and box dimensions.

No offline augmented copies were added to the validation or internal test subsets. During training only, mosaic augmentation, random affine transformations, horizontal flipping, and hue–saturation–value (HSV) jittering were applied online to training batches [10]. These operations altered the appearance of training samples without changing the stored dataset partitions, thereby avoiding augmentation-derived overlap between the training and evaluation subsets [19].

2.2.2. Dataset Characteristics and Partitioning

After manual screening removed severely corrupted or completely out-of-focus captures, the curated dataset contained 1998 images. The approximate image ratio for Kiss Tomato, Millennium Tomato, and White Jade Tomato was 2:1:1, respectively, reflecting image availability at the collection site. The images were resized from 3072 × 4096 pixels to 640 × 640 pixels for model input.

As illustrated in Figure 2, the three cultivars differ in fruit color, surface appearance, cluster density, and surrounding canopy. Millennium tomatoes often appear in compact clusters, increasing overlap between neighboring boxes. White Jade tomatoes have a pale, low-saturation appearance that can reduce contrast against foliage. White Jade is therefore discussed as a visually low-contrast detection target rather than in terms of unmeasured cultivar-specific biochemical properties.

Before annotation, related photographs and continuous views of the same plant were grouped and assigned to one subset only. The groups were then allocated at approximately 8:1:1 to the training, validation, and internal held-out test subsets, respectively, as detailed in Table 1 [20]. The validation subset supported training monitoring and final weight selection, whereas the internal test subset was reserved for evaluation after model and hyperparameter selection. Because all subsets came from the same orchard, the test subset

provides internal evaluation rather than independent external validation. The manual annotations comprise 9162 labeled fruit instances (Table 2); these counts do not represent sampled plants or horticultural replications.



(a) Kiss Tomato (b) Millennium Tomato (c) White Jade Tomato

(d) High Illumination (e) Shadow/Backlight (f) Complex Background

(g) Severe Occlusion (h) Dense Clustering (i) Extreme Scale Variation

Figure 2. Representative handheld mobile phone RGB images from the custom orchard dataset rather than examples from a fixed acquisition platform. The panels show (**a**) densely clustered Kiss Tomatoes; (**b**) clustered Millennium Tomatoes; (**c**) low-saturation White Jade Tomatoes; (**d**) exposure highlights under direct illumination; (**e**) canopy shadow and localized backlighting; (**f**) complex background foliage; (**g**) leaf occlusion; (**h**) overlapping fruit clusters; and (**i**) variation in target scale.

Table 1. Image-level partitioning of the multi-class tomato dataset into the training, validation, and internal test subsets. The listed resolution denotes the network input size after resizing rather than the original acquisition resolution.

Dataset Split	Number of Images	Ratio (%)	Input Resolution (Pixels)
Training Subset	1598	80%	640 × 640
Validation Subset	200	10%	640 × 640
Internal Test Subset	200	10%	640 × 640
Total	1998	100%	–

Table 2. Distribution of manually annotated tomato instances across varieties and dataset subsets.

Category	Training	Validation	Internal Test	Total Instances
Kiss Tomato	2987	464	555	4006
Millennium Tomato	1749	253	272	2274
White Jade Tomato	2337	282	263	2882
Total Instances	7073	999	1090	9162

2.2.3. Implementation Details and Hyperparameter Protocol

Experiments were conducted in a cloud Linux container with an Intel Core i9 CPU and an NVIDIA RTX 4080-class virtual GPU with 32 GB of allocated memory. The models were implemented in PyTorch 2.0.

All models were trained from random initialization for 300 epochs to exclude differences caused by external domain pre-training and to compare the architectures under the same target-domain setting. A batch size of 128 was used. Stochastic gradient descent (SGD) was applied with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. Online augmentation, including mosaic augmentation, random affine transformations, horizontal flipping, and HSV color jittering, was applied only to the training subset [21].

For the stepwise ablation study in Table 3, each of the eight configurations was trained independently with random seeds 0, 1, 2, 3, and 4 under otherwise identical settings. The table reports the mean and sample standard deviation (SD) across the five runs. Because the configurations were evaluated repeatedly under matched seeds, each detection metric was first compared across configurations using a Friedman test. A targeted Nemenyi post-hoc comparison was then made between WCAF-YOLO and the YOLOv26n baseline. Holm correction was applied separately across the four Friedman p-values and across the four Nemenyi p-values, with a two-sided significance level of 0.05. Statistical calculations were implemented in Python 3.12.3 using SciPy 1.13.0. The component-level and benchmark comparisons in the remaining tables retain the original seed-0 protocol and are reported as single-run results.

Table 3. Stepwise ablation analysis of the proposed architectural enhancements on the multi-variety tomato dataset. For each detection metric, the first line reports the mean, and the parenthesized value below reports the sample standard deviation (SD) across five runs with random seeds 0–4. The final configuration and the highest mean detection metric in each column are shown in bold. Checkmarks (✓) indicate the integration of the corresponding component, while a dash (-) indicates that the module was omitted.

Model	Components			Detection Performance				Complexity	
	WCAF	SPD	Loss	*P*	*R*	mAP_{50}	$mAP_{50–95}$	Params (M)	GFLOPs
Baseline (YOLOv26n)	-	-	-	0.9422 (0.0031)	0.8938 (0.0024)	0.9540 (0.0020)	0.8834 (0.0027)	2.38	5.19
Model-A	✓	-	-	0.9562 (0.0016)	0.8980 (0.0032)	0.9604 (0.0017)	0.8866 (0.0023)	2.00	4.74
Model-B	-	✓	-	0.9508 (0.0022)	0.9046 (0.0017)	0.9650 (0.0021)	0.8880 (0.0020)	2.38	6.46
Model-C	-	-	✓	0.9556 (0.0005)	0.9022 (0.0019)	0.9606 (0.0013)	0.8886 (0.0011)	2.38	5.19
Model-D	✓	✓	-	0.9572 (0.0008)	0.8974 (0.0019)	0.9612 (0.0008)	0.8856 (0.0017)	2.36	6.36
Model-E	✓	-	✓	0.9580 (0.0016)	0.8980 (0.0019)	0.9604 (0.0017)	0.8842 (0.0018)	2.00	4.74
Model-F	-	✓	✓	0.9552 (0.0031)	0.9070 (0.0017)	0.9642 (0.0018)	0.8880 (0.0026)	2.38	6.46
WCAF-YOLO (Final)	✓	✓	✓	**0.9644** (0.0015)	**0.9280** (0.0019)	**0.9716** (0.0013)	**0.9048** (0.0013)	2.36	6.36

2.2.4. Evaluation Metrics

Detection accuracy was evaluated on the internal held-out test subset using precision (P), recall (R), and mean average precision (mAP). Precision and recall are defined as

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \tag{1}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{2}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. The recall metric is particularly relevant in dense canopy images because it reflects the frequency of missed detections.

Average precision (AP) is the area under the precision-recall curve for one category, and mAP is the unweighted mean of AP across the $N = 3$ categories:

$$\mathrm{AP} = \int_0^1 P(R)\, dR, \tag{3}$$

$$\mathrm{mAP} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{AP}_i. \tag{4}$$

We report mAP_{50} at an intersection over union threshold of 0.5 and $\mathrm{mAP}_{50\text{–}95}$ averaged over thresholds from 0.50 to 0.95 in increments of 0.05. Model complexity is reported as the number of parameters (Params, in millions), giga floating-point operations (GFLOPs), and network layers.

2.3. SPD-Conv for Spatial-Preserving Downsampling

Standard object detectors often use strided convolutions or max pooling to reduce feature map dimensions and expand the receptive field. These operations can weaken fine spatial detail in precision-agriculture images [22]. In dense tomato canopies, small or distant targets may occupy few pixels. A stride of 2 halves each spatial dimension and produces one quarter as many output locations at the next scale. This reduction can make boundary representation more difficult under heavy leaf occlusion [23].

To address this bottleneck, we integrate a space-to-depth (SPD) convolutional block into the baseline topology [17]. Rather than reducing spatial resolution only through decimation, the SPD module uses deterministic tensor reshaping to transfer sampled spatial positions into the channel dimension. For a feature tensor X with height H, width W, and C channels, the operation forms four sub-tensors:

$$\begin{aligned} X_{0,0} &= X[0:H:2, 0:W:2, :] \\ X_{1,0} &= X[1:H:2, 0:W:2, :] \\ X_{0,1} &= X[0:H:2, 1:W:2, :] \\ X_{1,1} &= X[1:H:2, 1:W:2, :]. \end{aligned} \tag{5}$$

The four sub-tensors are concatenated along the channel axis to produce

$$X_{\mathrm{spd}} \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times 4C}.$$

This rearrangement transfers sampled spatial positions into the channel dimension while reducing spatial resolution. Figure 3 illustrates the transformation.

Following this spatial-to-depth transformation, a non-strided convolution (stride of 1) is applied. The final output tensor is

$$X_{\text{out}} \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times C_{\text{out}}}.$$

This operation compresses the expanded channel dimension to the target depth while limiting the loss of sampled spatial information.

Within the WCAF-YOLO backbone, the SPD-Conv module is placed at the early P3 feature extraction stage, where spatial details of small or obscured tomatoes can be affected by downsampling. Replacing the conventional downsampling block at this stage is intended to pass more boundary-related cues into the deeper feature pyramid. Consequently, the network can retain more localization cues for overlapping canopy fruits without requiring ultra-high-resolution inputs throughout the full network.

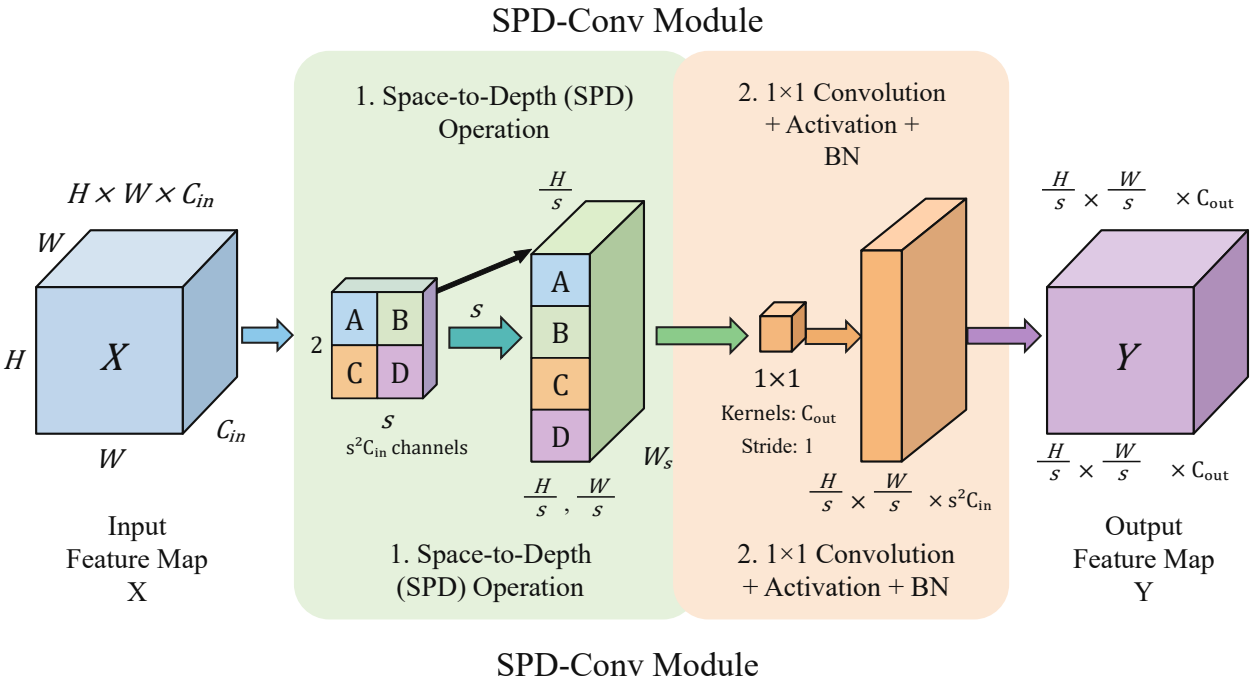


Figure 3. Volumetric illustration of the spatial-to-depth mechanism in SPD-Conv. *H*, *W*, and *C* denote feature-map height, width, and channel count, respectively. The colored A–D blocks identify the four spatial offsets that are sliced and stacked along the channel axis, increasing the depth to 4C; arrows indicate forward tensor flow. A non-strided 1 × 1 convolution then compresses the expanded channel dimension to C_{out}. The schematic was independently redrawn by the authors from the operation described in [17]; no third-party graphical elements were reused.

2.4. Weighted Channel-Aware Fusion (WCAF) Network

Conventional feature pyramid architectures, such as PANet [24], combine multi-scale feature maps through concatenation or element-wise addition without explicit branch-level weights at the fusion operation. In orchard scenes, shallow maps may retain foliage texture and chromatic highlights, whereas deeper maps contain stronger semantic information but less spatial detail. An explicit weighting step may help regulate these heterogeneous inputs before they reach the detection heads.

To reduce this feature misalignment, we designed the weighted channel-aware fusion (WCAF) module. WCAF replaces selected concatenation nodes in the neck architecture with a lightweight dynamic calibration structure. As shown in Figure 4, the operation includes two sequential stages: learnable tensor weighting and a parameter-free 3D energy-based attention refinement mechanism.

In the initial stage, WCAF uses a small set of learnable scalar weights to regulate the contribution of each incoming branch. For two input tensors, X_0 and X_1, WCAF learns the scalar branch weights w_0 and w_1. To maintain non-negativity and numerical stability during backpropagation, the normalized weighted fusion is formulated as [25]:

$$X_{\text{fused}} = \text{Concat}\left(X_0 \cdot \frac{\text{ReLU}(w_0)}{\sum_{i=0}^{1} \text{ReLU}(w_i) + \epsilon}, \; X_1 \cdot \frac{\text{ReLU}(w_1)}{\sum_{i=0}^{1} \text{ReLU}(w_i) + \epsilon} \right), \tag{6}$$

where $\epsilon = 10^{-4}$ prevents division by zero, and ReLU rectifies negative weight values. The normalized weights allow the relative contribution of each branch to be learned during optimization.

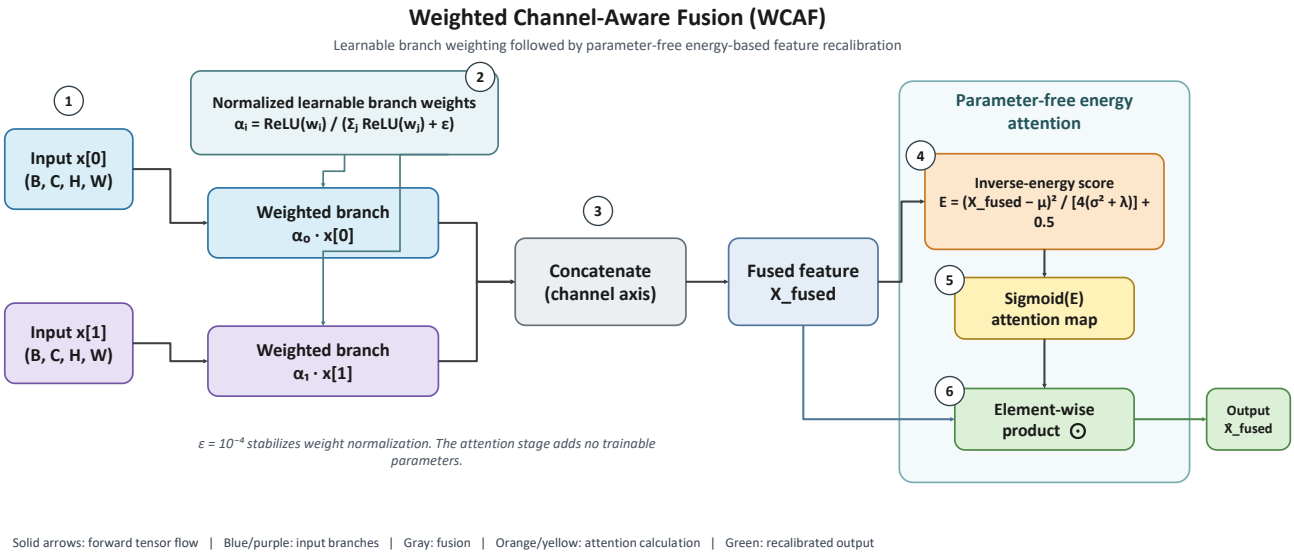


Figure 4. Architecture of the weighted channel-aware fusion (WCAF) module. The two input tensors, X_0 and X_1, are scaled using normalized non-negative learnable weights and concatenated along the channel dimension. A parameter-free inverse-energy function then produces a sigmoid attention map, which recalibrates the fused feature through element-wise multiplication ($\odot$). Numbered labels identify the processing stages, and solid arrows denote forward tensor flow. Blue and purple identify the input branches, gray denotes feature fusion, orange and yellow denote attention calculation, and green denotes the recalibrated output; the colors distinguish functional stages rather than numerical values.

Although weighted concatenation adjusts the contribution of each scale, the resulting tensor $X_{\text{fused}} \in \mathbb{R}^{C \times H \times W}$ may still contain locally redundant responses. Instead of an attention module with additional trainable layers, WCAF applies a parameter-free three-dimensional energy-based refinement to the fused tensor.

Following SimAM [26], we calculate the spatial mean μ_c and variance σ_c^2 of each channel over dimensions H and W. The resulting inverse-energy score E estimates the relative importance of each activation without additional learnable parameters:

$$E_{c,i,j} = \frac{(X_{\text{fused}}(c,i,j) - \mu_c)^2}{4(\sigma_c^2 + \lambda)} + 0.5, \tag{7}$$

where $\lambda = 10^{-4}$ is a regularization constant. In this expression, a larger deviation from the channel mean produces a larger inverse-energy score and, after sigmoid activation, a larger weight. This operation is intended to emphasize locally distinctive responses. Its relevance

to tomato contours remains a feature-level interpretation rather than direct evidence of a causal mechanism.

Finally, the synthesized energy map is projected through a sigmoid activation to generate 3D attention weights, which modulate the fused feature map:

$$\hat{X}_{\text{fused}} = X_{\text{fused}} \odot \text{Sigmoid}(E), \tag{8}$$

where $\odot$ denotes element-wise multiplication, also known as the Hadamard product [27]. The branch weights add only a small number of trainable scalars, while the subsequent attention refinement is parameter-free.

2.5. Focaler-MPDIoU Loss for Dense Target Regression

Conventional bounding box regression metrics, such as complete intersection over union (CIoU) [28] implemented in the baseline YOLO architecture, primarily penalize the Euclidean distance between geometric centers and the variance in aspect ratios. Although computationally stable for isolated large objects, this formulation can be less discriminative in dense tomato canopies. In clustered environments, highly overlapping fruits may generate bounding boxes with similar aspect ratios and nearby center coordinates. This geometric redundancy can weaken the distinction between neighboring targets and contribute to false negatives for severely occluded fruits.

To improve bounding box regression under dense occlusion, we substitute the baseline metric with minimum point distance intersection over union (MPDIoU) [29]. MPDIoU uses the distances between corresponding corners of the predicted box (B_p) and ground-truth box (B_g). Let (x_1^p, y_1^p) and (x_2^p, y_2^p) denote the top-left and bottom-right coordinates of B_p, while (x_1^g, y_1^g) and (x_2^g, y_2^g) denote those of B_g. The metric is formulated as

$$\text{MPDIoU} = \text{IoU} - \frac{d_{tl}^2}{w_c^2 + h_c^2} - \frac{d_{br}^2}{w_c^2 + h_c^2}, \tag{9}$$

where d_{tl}^2 and d_{br}^2 are the squared Euclidean distances between the corresponding top-left and bottom-right corners. The variables w_c and h_c denote the width and height of the smallest rectangle enclosing both boxes.

Although MPDIoU provides a geometric formulation for box alignment, training samples can still contribute unequally to optimization. We therefore integrate the Focaler mechanism [18], which applies a bounded piecewise linear mapping. With u and d denoting the upper and lower thresholds, respectively, the mapped metric is formulated as

$$\text{MPDIoU}^{\text{focaler}} = \begin{cases} 0, & \text{MPDIoU} < d \\ \frac{\text{MPDIoU} - d}{u - d}, & d \le \text{MPDIoU} \le u \\ 1, & \text{MPDIoU} > u \end{cases}. \tag{10}$$

The final bounding box regression loss is consequently defined as

$$L_{\text{Focaler-MPDIoU}} = 1 - \text{MPDIoU}^{\text{focaler}}. \tag{11}$$

The combination of these two mechanisms is illustrated in Figure 5. The final WCAF-YOLO configuration adopts the Focaler-MPDIoU loss during training; it does not introduce additional parameters or GFLOPs during inference. In the ablation experiments, the configuration using Focaler-MPDIoU obtained higher recall and $mAP_{50–95}$ values than the corresponding alternatives under the same protocol.

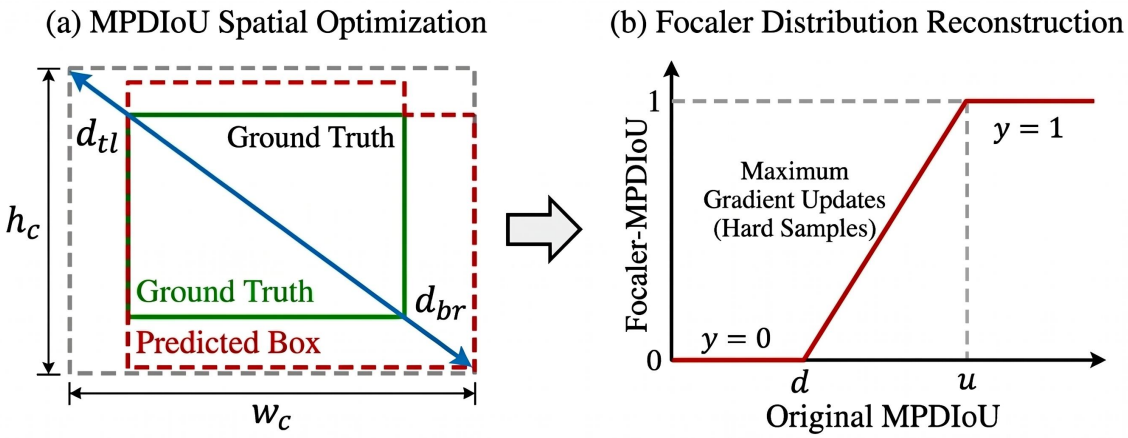


Figure 5. Conceptual diagram of the Focaler-MPDIoU mechanism. (**a**) The green solid box denotes the ground-truth box, the red dashed box denotes the predicted box, and the gray dashed rectangle denotes the smallest enclosing box. Blue diagonals indicate the corner distances d_{tl} and d_{br}, while black dimension arrows indicate w_c and h_c. (**b**) The red solid line represents the focaler piecewise mapping, and the gray dashed guides mark the lower and upper thresholds d and u. Colors and line styles distinguish geometric quantities and do not encode measured values.

3. Results

This section presents the ablation experiments, component-level comparisons, qualitative visual results, and benchmarking against representative detectors. All results use the dataset partitions, training protocol, and evaluation metrics defined in Section 2.2.

3.1. Ablation Study on Proposed Modules

To evaluate the individual and combined contributions of the SPD-Conv downsampling mechanism, the weighted channel-aware fusion (WCAF) module, and the Focaler-MPDIoU loss, we conducted a sequence of ablation experiments. Using the standard YOLOv26n as the baseline, each component was incrementally integrated to form the final WCAF-YOLO structure. The impact of these modifications on detection metrics and architectural complexity is detailed in Table 3.

The ablation results in Table 3 show different metric and complexity changes for the three proposed components. When only the WCAF module is introduced, Model-A obtains higher mean precision, recall, mAP_{50}, and $mAP_{50–95}$ than the YOLOv26n baseline, while reducing the parameter count from 2.38 M to 2.00 M and the computational cost from 5.19 to 4.74 GFLOPs. This result refers specifically to the WCAF-only configuration rather than the final model. The compactness of this configuration is associated with the lightweight fusion design, which combines channel adjustment with parameter-free energy-based refinement. Introducing SPD-Conv increases GFLOPs but is also associated with higher mean recall and mAP_{50} in the corresponding ablation setting.

The final WCAF-YOLO configuration integrates WCAF, SPD-Conv, and Focaler-MPDIoU and obtained the highest mean value for each detection metric among the tested ablation settings. Relative to the YOLOv26n baseline, the mean gains were 2.22 percentage points in precision, 3.42 in recall, 1.76 in mAP_{50}, and 2.14 in $mAP_{50–95}$. The Friedman tests detected differences among the eight configurations for precision ($\chi^2(7) = 29.43$), recall ($\chi^2(7) = 32.82$), mAP_{50} ($\chi^2(7) = 29.88$), and $mAP_{50–95}$ ($\chi^2(7) = 30.20$); all four Holm-adjusted p-values were below 0.001. In the targeted Nemenyi post-hoc comparisons, WCAF-YOLO also differed from the baseline for all four metrics after Holm correction ($p < 0.001$). These repeated-run results support the combined design under the evaluated protocol, but they do not establish independent causal effects for each component. The final model contains 2.36 M parameters and requires 6.36 GFLOPs.

3.2. In-Depth Comparative Analysis of Key Components

This subsection further examines the two main design choices that directly affect feature aggregation and bounding box regression. The comparisons are conducted under controlled settings so that differences can be attributed to the evaluated component.

3.2.1. Performance of the Proposed WCAF Neck

To evaluate the proposed weighted channel-aware fusion (WCAF) module, we conducted a comparative experiment. Using a YOLOv26n-SPD base with Focaler-MPDIoU as the intermediate framework, we compared WCAF with BiFPN [25] and five attention mechanisms: squeeze-and-excitation (SE) [30], convolutional block attention module (CBAM) [31], efficient channel attention (ECA) [32], coordinate attention (CA) [33], and efficient multi-scale attention (EMA) [34]. The results are detailed in Table 4.

As shown in Table 4, replacing the standard neck with BiFPN reduced the parameter count but slightly decreased $mAP_{50–95}$ from 0.887 to 0.883. Thus, the smaller BiFPN configuration did not improve detection accuracy under this protocol.

The SE, CBAM, and ECA configurations produced similar or lower $mAP_{50–95}$ values than the intermediate baseline. CA and EMA increased $mAP_{50–95}$ to 0.892 and 0.889, respectively, but also increased computation. EMA required 8.39 GFLOPs. These comparisons describe performance in the present architecture and dataset; they do not establish a general ranking of attention mechanisms.

Table 4. Performance comparison of various attention mechanisms and neck architectures. Note: All models evaluated in this table are built upon the YOLOv26n-SPD architecture and trained using our proposed Focaler-MPDIoU loss. This setting helps isolate the performance variations associated with different neck configurations, and the best detection metrics are highlighted in bold.

Method	*P*	*R*	mAP_{50}	$mAP_{50–95}$	Params (M)	GFLOPs	Layers
Baseline (SPD + Proposed Loss)	0.955	0.907	0.964	0.887	2.38	6.46	241
+BiFPN Neck	0.940	0.895	0.953	0.883	1.74	5.59	232
+SE Attention	0.947	0.907	0.964	0.886	2.43	6.51	269
+ECA Attention	0.947	0.905	0.964	0.888	2.38	6.47	257
+CBAM Attention	0.955	0.901	0.958	0.885	2.43	6.55	285
+CA Attention	0.953	0.897	0.965	0.892	2.42	6.56	281
+EMA Attention	0.957	0.911	0.959	0.889	2.45	8.39	273
+WCAF (WCAF-YOLO)	**0.964**	**0.929**	**0.971**	**0.904**	2.36	6.36	245

Among the evaluated alternatives, WCAF achieved the highest precision (96.4%), recall (92.9%), mAP_{50} (0.971), and $mAP_{50–95}$ (0.904). Relative to the intermediate baseline, its parameter count decreased from 2.38 M to 2.36 M and its cost decreased from 6.46 to 6.36 GFLOPs. This comparison is limited to the tested neck configurations and dataset.

3.2.2. Evaluation of the Bounding Box Regression Loss

To compare bounding box regression strategies for heavily occluded agricultural scenes, we evaluated several loss functions. Keeping the structural architecture constant (YOLOv26n-SPD integrated with the WCAF neck), we compared Focaler-MPDIoU with the standard CIoU and six recently proposed metrics: EIoU [35], SIoU [36], Shape-IoU [37], Inner-IoU [38], WIoU [39], and MPDIoU [29]. Modifying the loss function affects training time optimization and introduces no additional parameters during inference. The results are detailed in Table 5.

Table 5. Comparative performance of various bounding box regression loss functions integrated into our architectural baseline. All configurations maintain identical computational complexity (2.36 M parameters, 6.36 GFLOPs, and 245 layers). The best results are highlighted in bold.

Loss Function	*P*	*R*	mAP_{50}	$mAP_{50–95}$
CIoU (Baseline)	0.958	0.898	0.962	0.884
EIoU	0.948	0.899	0.952	0.883
SIoU	0.945	0.901	0.958	0.884
Shape-IoU	0.952	0.907	0.962	0.892
Inner-IoU	0.959	0.902	0.959	0.893
WIoU	0.954	0.907	0.961	0.894
MPDIoU	0.960	0.911	0.966	0.895
Focaler-MPDIoU (WCAF-YOLO)	**0.964**	**0.929**	**0.971**	**0.904**

Table 5 shows that the regression losses produced different results under the same architecture. EIoU and SIoU did not improve $mAP_{50–95}$ over the CIoU baseline in this experiment. Because these objectives differ in several geometric terms, the comparison does not isolate a single cause for their performance.

Shape-IoU and Inner-IoU achieved $mAP_{50–95}$ scores of 0.892 and 0.893, respectively. WIoU reached 0.894, and MPDIoU produced the highest pre-Focaler value of 0.895. These results show that the corner distance formulation was competitive among the tested regression objectives, without by itself proving why the improvement occurred.

The final WCAF-YOLO configuration combines the Focaler mapping with MPDIoU. Focaler-MPDIoU achieved the highest $mAP_{50–95}$ of 0.904 and recall of 0.929. The recall increased by 1.8 percentage points over MPDIoU and by 3.1 percentage points over CIoU. The result supports the use of the combined objective in this model, while the experiment does not directly measure per-sample gradient contributions.

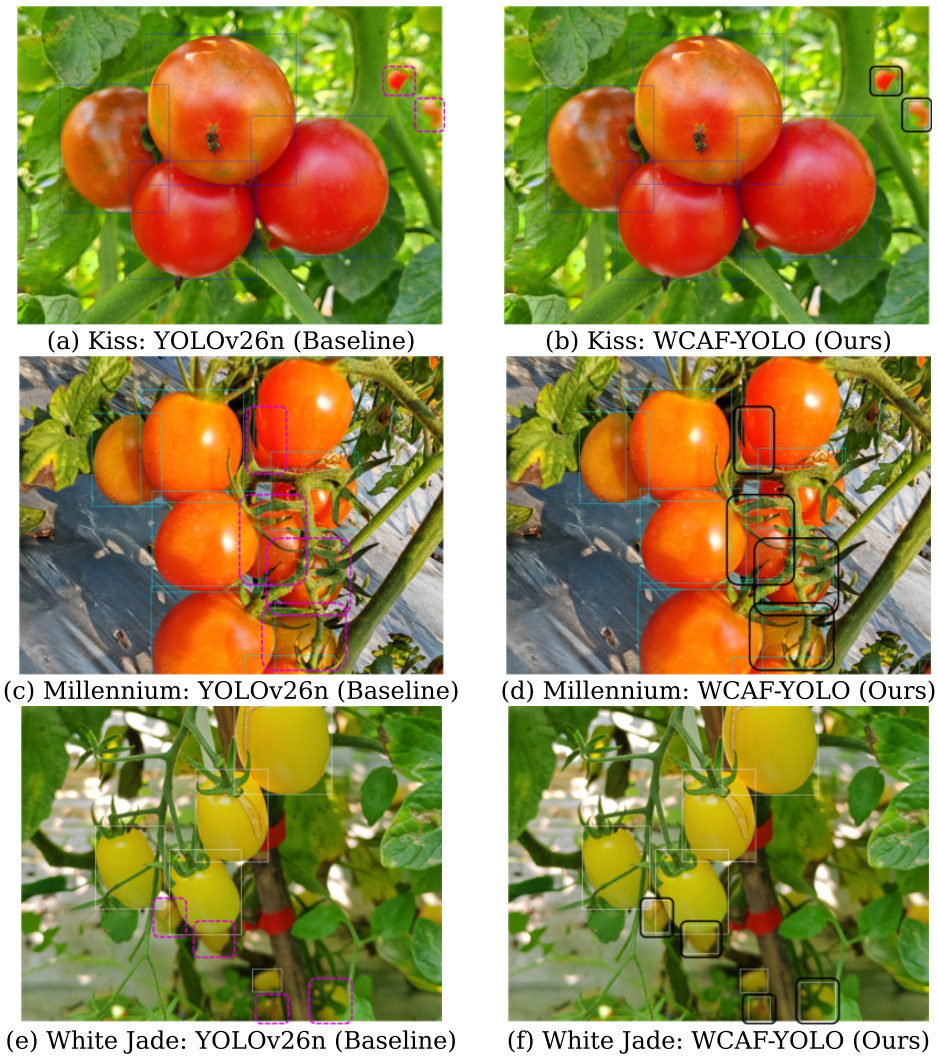
3.2.3. Qualitative Results and Visual Comparison

To further examine the detection behavior of WCAF-YOLO in complex orchard images, we conducted a qualitative visual comparison against the YOLOv26n baseline. Representative samples from the internal test subset, covering three tomato cultivars under dense clustering, leaf occlusion, and background interference, are illustrated in Figure 6. These examples show image-level detection outputs and should not be interpreted as a specific camera mount, robot platform, or acquisition system.

The visual comparison illustrates several representative differences between the baseline and WCAF-YOLO:

- **Dense clustering and small peripheral fruit (Kiss Tomato):** As shown in the first row of Figure 6, the baseline model misses small distant fruits near a dense foreground cluster. In the corresponding WCAF-YOLO result, these peripheral fruits are detected as separate instances, which is consistent with the recall improvement observed in the quantitative results.
- **Severe occlusion (Millennium Tomato):** In the second row, the Millennium tomatoes are partially obscured by foliage and stems. The baseline model shows localization drift in several highlighted regions, whereas WCAF-YOLO produces bounding boxes that better align with the visible fruit regions.
- **Low-contrast background (White Jade Tomato):** The third row shows detection under conditions where White Jade tomatoes have relatively low color contrast with the surrounding foliage. Compared with the baseline, WCAF-YOLO detects additional small or partially occluded fruits in low-contrast regions, which may be related to improved retention of fine-grained spatial information.

In these selected examples, WCAF-YOLO showed fewer missed or poorly localized fruit than the baseline. This qualitative comparison is consistent with the higher recall on the complete internal test subset.



(a) Kiss: YOLOv26n (Baseline) (b) Kiss: WCAF-YOLO (Ours)

(c) Millennium: YOLOv26n (Baseline) (d) Millennium: WCAF-YOLO (Ours)

(e) White Jade: YOLOv26n (Baseline) (f) White Jade: WCAF-YOLO (Ours)

Figure 6. Visual comparison between the YOLOv26n baseline and WCAF-YOLO for three tomato varieties. The panels are representative examples from the internal test subset, not images from a fixed acquisition platform. Blue boxes are the original model predictions. Auxiliary callouts were added after inference to aid comparison at normal magnification and are not detector outputs. Magenta dashed callouts mark regions where the baseline produced no separate box or only partial localization. Black solid callouts mark the corresponding regions detected by WCAF-YOLO.

3.3. Quantitative Comparison with Representative Detectors

To evaluate WCAF-YOLO beyond the YOLOv26n baseline, we compared it with detectors from different architectural families. The comparison included Faster R-CNN as a two-stage detector, RT-DETR-R18 as a transformer-based end-to-end detector, and lightweight YOLO-family models as single-stage baselines [10,40–42]. Faster R-CNN used a ResNet-50-FPN backbone and is reported as "Faster R-CNN" in Table 6. All models were trained from random initialization using the same data split, input resolution, number of epochs, hardware environment, and hyperparameter settings. This matched protocol excludes differences due to external pre-training, although it does not constitute model-specific hyperparameter optimization.

The two non-YOLO baselines provide useful references for model capacity and detection accuracy. Faster R-CNN and RT-DETR-R18 achieved strong $mAP_{50–95}$ scores of 0.896 and 0.892, respectively, confirming their ability to localize tomatoes in complex scenes. However, this accuracy was accompanied by substantially higher computational costs: Faster R-CNN required 41.53 M parameters and 206.00 GFLOPs, while RT-DETR-R18 required 20.00 M parameters and 60.00 GFLOPs. Such computational budgets are considerably higher than those of the lightweight YOLO-family models and may limit their suitability for resource-constrained agricultural edge platforms. These results suggest that simply increasing model capacity is not the most efficient strategy for addressing occlusion, clustering, and background interference in orchard environments.

Table 6. Comprehensive performance benchmarking of the proposed WCAF-YOLO against representative detection models. All models were trained from scratch under the same data split, input resolution, training epochs, hardware environment, and hyperparameter settings. In this table, Faster R-CNN denotes the Faster R-CNN detector with a ResNet-50-FPN backbone. Values are single-run results obtained using the matched seed-0 protocol; statistical inference is restricted to the repeated-run ablation study in Table 3. The best detection metrics and the lowest Params/GFLOPs are highlighted in **bold**.

Model	Params (M)	GFLOPs	P	R	mAP_{50}	$mAP_{50–95}$	Layers
Faster R-CNN	41.53	206.00	0.959	0.911	0.968	0.896	335
RT-DETR-R18	20.00	60.00	0.957	0.912	0.966	0.892	280
YOLOv5n	2.50	7.07	0.948	0.864	0.956	0.864	194
YOLOv6n	4.23	11.78	0.949	0.901	0.965	0.894	143
YOLOv8n	3.01	8.09	0.937	0.899	0.961	0.878	169
YOLOv9t	**1.97**	7.60	0.943	0.869	0.951	0.867	486
YOLOv10n	2.27	6.53	0.941	0.857	0.943	0.854	230
YOLO11n	2.58	6.32	0.945	0.904	0.964	0.889	239
YOLOv12n	2.56	6.32	0.949	0.885	0.953	0.889	352
YOLO13n	2.45	6.20	0.955	0.903	0.962	0.892	535
YOLOv26n (Base)	2.38	**5.19**	0.945	0.895	0.953	0.883	284
WCAF-YOLO (Ours)	2.36	6.36	**0.964**	**0.929**	**0.971**	**0.904**	245

Among the lightweight YOLO-family baselines, YOLOv6n achieved an $mAP_{50–95}$ of 0.894 with 4.23 M parameters and 11.78 GFLOPs. The more compact YOLOv9t and YOLOv10n configurations had lower recall, whereas YOLOv12n and YOLO13n reached $mAP_{50–95}$ values of 0.889 and 0.892. Thus, parameter count alone did not predict detection accuracy under the matched protocol.

WCAF-YOLO achieved the highest overall detection metrics among the evaluated models, with a precision of 0.964, recall of 0.929, mAP_{50} of 0.971, and $mAP_{50–95}$ of 0.904. Compared with the YOLOv26n baseline, it improved recall by 3.4 percentage points and $mAP_{50–95}$ by 2.1 percentage points, while maintaining a slightly smaller parameter count. Compared with Faster R-CNN, WCAF-YOLO improved $mAP_{50–95}$ by 0.8 percentage points and recall by 1.8 percentage points, while using 2.36 M rather than 41.53 M parameters. Under the tested conditions, the final model combined higher detection metrics with a substantially smaller parameter count than the two-stage and transformer baselines.

Figure 7 places the benchmarking results in the accuracy–complexity plane. WCAF-YOLO lies near the upper-left region because it combines the highest $mAP_{50–95}$ in the comparison with a parameter count of 2.36 M, although it does not have the lowest GFLOPs.

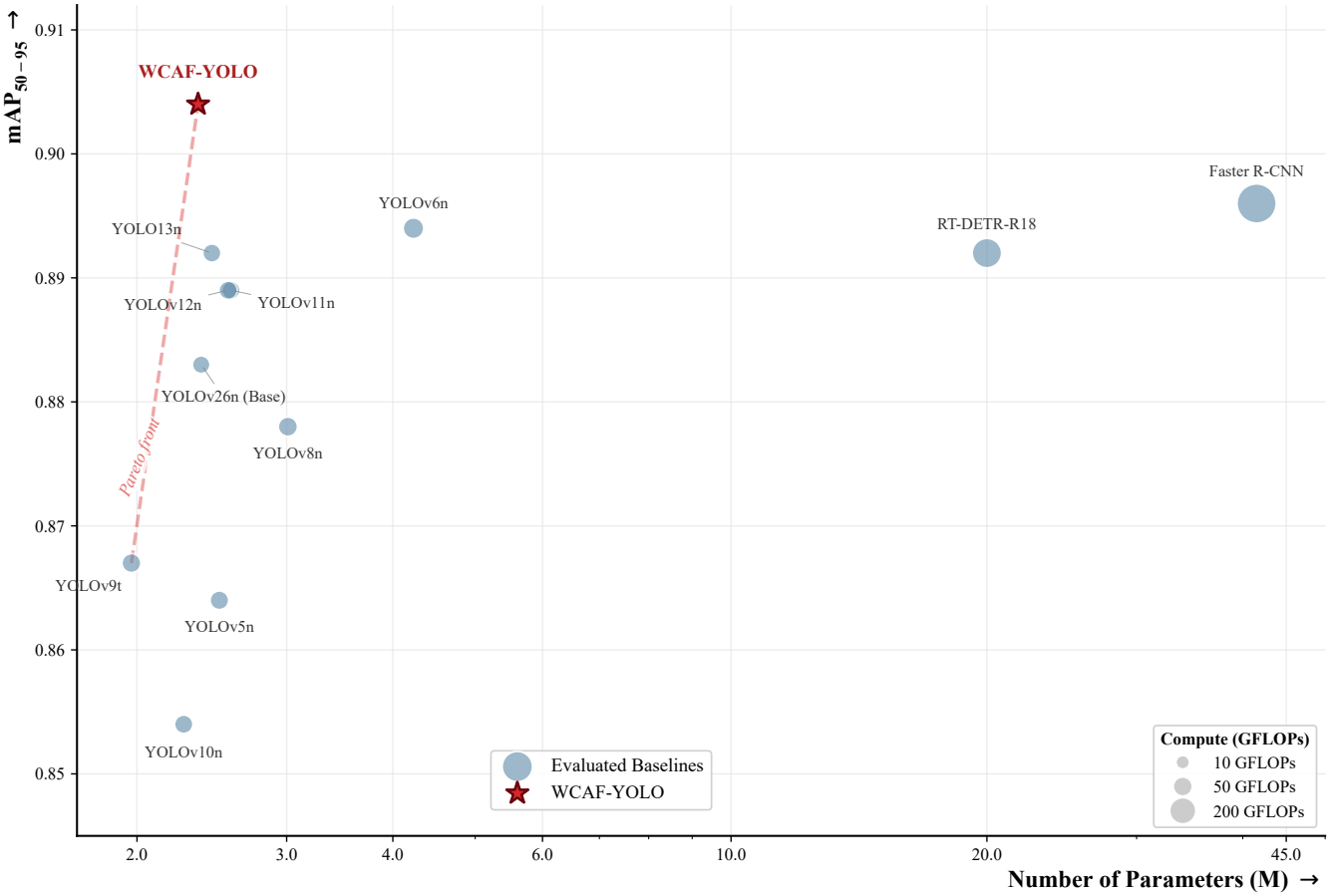


Figure 7. Accuracy–complexity comparison of the evaluated detectors. Blue circles denote the evaluated baselines, the red star denotes WCAF-YOLO, and marker area represents GFLOPs. The dashed red segment connects the non-dominated models in the parameter–accuracy plane. Horizontal and vertical axis arrows indicate increasing parameter count and $mAP_{50–95}$, respectively; thin gray leader lines, where present, link labels to crowded data points. WCAF-YOLO is located near the top-left region because it combines the highest $mAP_{50–95}$ in this comparison with 2.36 M parameters; it does not have the lowest GFLOPs.

4. Discussion

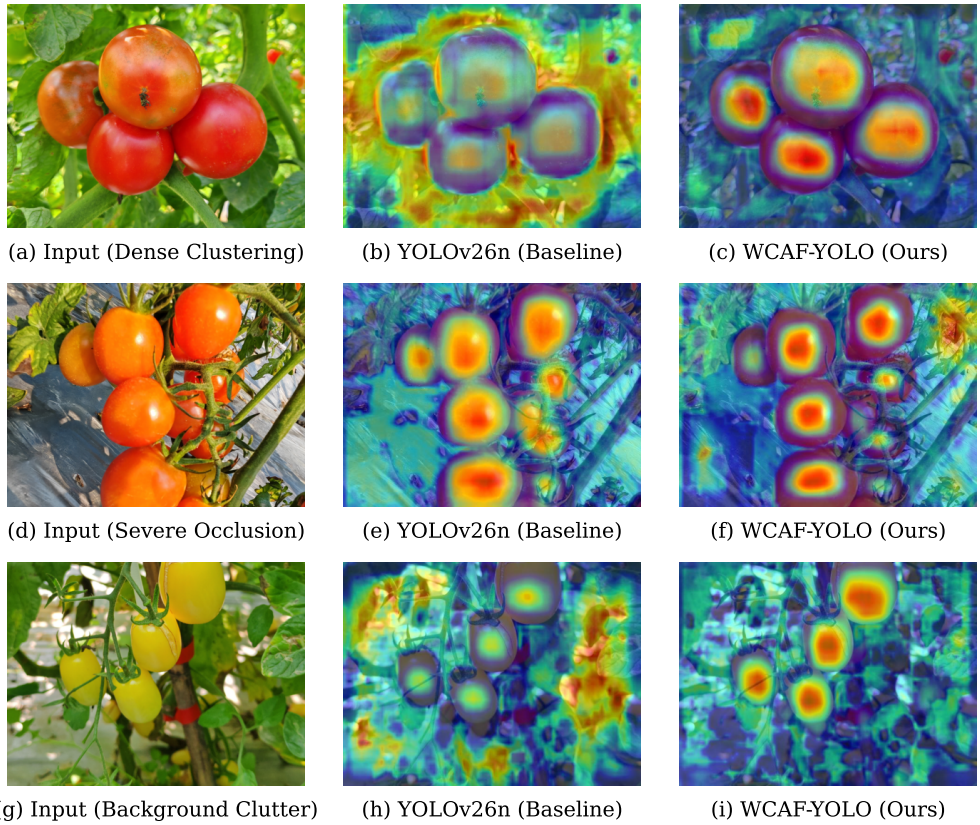
The following discussion considers the activation maps, implications for future edge deployment, and limits of the current evidence. It separates observations made in this study from applications that still require validation.

4.1. Activation-Based Interpretation of the Lightweight Architecture

While quantitative metrics (e.g., mAP and recall) provide an overall evaluation of model performance, they do not directly show how image regions contribute to predictions. To obtain an auxiliary visualization of model behavior, we employed Eigen-CAM [43] to map the spatial distribution of convolutional activations. This technique projects high-level activation patterns onto the original image space and provides a qualitative reference for comparing the baseline and WCAF-YOLO under challenging orchard conditions.

The heatmaps in Figure 8 show more dispersed baseline activations in the selected dense-cluster and occlusion examples (Rows 1 and 2). Some responses extend to leaves and adjacent non-target regions, alongside the missed detections and localization errors shown in Figure 6. Because Eigen-CAM is qualitative, these examples do not identify which architectural component caused the pattern.

WCAF-YOLO shows more localized activation around tomato regions in the same examples. The difference is also visible for White Jade (Row 3), whose fruit boundaries have low contrast against surrounding leaves. This pattern is consistent with the higher recall of WCAF-YOLO, but it is not direct evidence of an internal causal mechanism.



(a) Input (Dense Clustering) (b) YOLOv26n (Baseline) (c) WCAF-YOLO (Ours)

(d) Input (Severe Occlusion) (e) YOLOv26n (Baseline) (f) WCAF-YOLO (Ours)

(g) Input (Background Clutter) (h) YOLOv26n (Baseline) (i) WCAF-YOLO (Ours)

Figure 8. Activation-based visualization using Eigen-CAM heatmaps. Representative challenging scenarios, including dense clustering (**Row 1**), foliage occlusion (**Row 2**), and background color clutter (**Row 3**), were selected to qualitatively compare the baseline and WCAF-YOLO. Within each heatmap, warm colors (yellow and red) indicate relatively stronger activation, whereas cool colors (blue and green) indicate relatively weaker activation. The colors are qualitative within each map and are not interpreted as causal evidence.

4.2. Implications for Future Edge Deployment

Although this study focuses on the visual perception algorithm, considering theoretical hardware requirements is useful for future integration into agricultural systems. Edge computing nodes embedded in mobile orchard monitoring devices or field perception modules—such as the NVIDIA Jetson series—typically operate under limited thermal and power budgets [16]. Consequently, parameter-heavy models such as Faster R-CNN or large transformer-based architectures may face deployment constraints related to thermal throttling, power consumption, and memory usage.

Within this operational envelope, WCAF-YOLO is a compact candidate for future edge-hardware evaluation. The model contains 2.36 M parameters and requires 6.36 GFLOPs, substantially less than the Faster R-CNN and RT-DETR-R18 baselines evaluated here. The energy-based attention stage is parameter-free, while branch fusion introduces only a small set of scalar weights. These complexity estimates motivate embedded-platform testing but do not establish real-device speed, energy use, or thermal performance. Such properties require direct evaluation on hardware such as NVIDIA Jetson devices [1,16].

4.3. Current Limitations and Future Work

Several limitations define the scope of the results. First, all images were collected from one commercial orchard in Baihe Town, Kaiping City. The validation and internal test subsets therefore share the collection domain and do not provide independent external validation. The results do not establish generalization across regions, cultivation systems, camera sensors, seasons, or independent datasets. Second, the collection was not designed as a plant-level replicated horticultural trial. Plant numbers, repetitions, images per plant, fruit height distributions, and physical fruit counts are unavailable. Future datasets should record these variables before image-level detections are linked to plant-level monitoring or yield estimation.

Third, the framework uses RGB input and does not treat unripe green tomatoes as a separate category. Visible and identifiable green fruits were assigned to their corresponding variety, but their similarity to foliage may reduce detection reliability. Because maturity-stage labels were not collected, performance for green fruits cannot be quantified separately. Reflections from pale White Jade surfaces and low illumination may also affect visible-light images. Future studies should use external orchards and multi-season data, add maturity-stage labels when required, and evaluate complementary sensing modalities such as RGB-D or other depth-based perception approaches [44].

Another limitation is the gap between two-dimensional (2D) object localization and physical three-dimensional (3D) actuation. The current detection framework outputs planar bounding boxes and variety labels, which are useful for image-level monitoring and sampling-based field assessment but do not provide the depth coordinates, pose, or approach vectors required for non-destructive manipulation. In dense fruit clusters, relying only on 2D bounding boxes may be insufficient for determining collision-free end-effector trajectories.

Future work should prioritize independent validation, plant-level monitoring protocols, and field-platform testing. Depth sensors, stereo vision, or point-cloud fusion could extend WCAF-YOLO toward 3D target localization and candidate manipulation poses [44,45]. The integration of depth sensing, tracking, grasp-point estimation, motion planning, and platform control remains outside the scope of this study. Tests on embedded hardware are also needed to measure inference speed, energy use, and detection stability under operational conditions.

5. Conclusions

This study investigated two-dimensional detection of Kiss, Millennium, and White Jade tomatoes in unstructured orchard images. Dense foliage, overlapping clusters, variable illumination, and background color interference make this task difficult. WCAF-YOLO addresses these conditions through targeted changes to downsampling, feature fusion, and box regression while retaining a compact computational profile.

The ablation results show that the proposed components were associated with the overall performance of WCAF-YOLO under the evaluated protocol. SPD-Conv was introduced to retain fine-grained spatial information during early feature downsampling, the weighted channel-aware fusion (WCAF) neck was designed to refine multi-scale feature fusion, and Focaler-MPDIoU was used for bounding box regression in densely clustered scenes. Across five random seed ablation runs, WCAF-YOLO obtained a mean $mAP_{50–95}$ of 0.9048 ± 0.0013 and a mean recall of 0.9280 ± 0.0019. The mean recall was 3.42 percentage points higher than that of the YOLOv26n baseline, and all four targeted post-hoc comparisons remained significant after Holm correction. In the original matched seed-0 benchmark, WCAF-YOLO recorded the highest precision, recall, mAP_{50}, and $mAP_{50–95}$ values among the evaluated two-stage, transformer-based, and lightweight one-stage detectors. The comparison set

was chosen to represent these common detector families and is not intended to establish performance beyond the single-site dataset or against every available architecture. The model maintained a compact structure with 2.36 M parameters and 6.36 GFLOPs.

Overall, WCAF-YOLO is a compact two-dimensional detector that outputs tomato boxes and variety labels in orchard images. Its parameter count and computational cost justify future evaluation on power-constrained edge devices. The present evidence is limited to image-level detection and does not validate fruit counting, yield estimation, maturity classification, 3D localization, or robotic harvesting. Further work should evaluate independent orchards and embedded platforms, collect plant-level and maturity-stage records, and test multi-source perception for 3D localization.

Author Contributions: Conceptualization, X.L. and Y.Z.; methodology, X.L. and Y.Z.; software, X.L., Y.Z. and L.Y.; validation, Z.D., B.W., Z.W. and Q.W.; formal analysis, X.L. and Y.Z.; investigation, X.L., Y.Z. and L.Y.; resources, Z.D. and X.T.; data curation, Y.Z. and Q.W.; writing—original draft preparation, X.L. and Y.Z.; writing—review and editing, X.T., Z.D. and B.W.; visualization, X.L. and Y.Z.; supervision, X.T. and Z.D.; project administration, X.T. and Z.D.; funding acquisition, X.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Science and Technology Project of Jiangmen (No. Jiangke [2025] 172), titled "Distinctive Integrated Agriculture, Culture and Tourism IP and Agricultural Development in Baihe Town".

Data Availability Statement: The data presented in this study are available from the corresponding author upon reasonable request. The data are not publicly available due to ongoing subsequent research and property restrictions.

Acknowledgments: We sincerely thank the cooperating commercial orchard and fruit growers in Baihe Town, Kaiping City, for granting access to the collection site and contributing practical knowledge during data collection and annotation review. We also thank the members of our research group at South China Agricultural University for their technical assistance.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CA	coordinate attention
CBAM	convolutional block attention module
CIoU	complete intersection over union
ECA	efficient channel attention
EMA	efficient multi-scale attention
FPN	feature pyramid network
GFLOPs	giga floating-point operations
HSV	hue-saturation-value
IoU	intersection over union
mAP	mean average precision
MPDIoU	minimum point distance intersection over union
RGB	red-green-blue
SE	squeeze-and-excitation
SGD	stochastic gradient descent
SPD	space-to-depth
SPPF	spatial pyramid pooling–fast
WCAF	weighted channel-aware fusion
YOLO	you only look once

References

1. Bakki, I.; Malyshev, D.; Carbone, G. Development of a Robotic Harvesting Device for Tomatoes Based on a Palm Gripper. In *Proceedings of I4SDG Workshop 2025—IFToMM for Sustainable Development Goals*; Carbone, G., Quaglia, G., Eds.; Mechanisms and Machine Science; Springer: Cham, Switzerland, 2025; Volume 180, pp. 229–238. [CrossRef]
2. Zhang, Y.; Wang, Z. Review of Robotic Grippers for High-Speed Handling of Fragile Foods. *Adv. Robot.* **2025**, *39*, 1054–1070. [CrossRef]
3. Ni, J.; Zhu, L.; Dong, L.; Wang, R.; Chen, K.; Gao, J.; Wang, W.; Zhou, L.; Zhao, B.; Rong, J.; et al. TomatoPoseNet: An Efficient Keypoint-Based 6D Pose Estimation Model for Non-Destructive Tomato Harvesting. *Agronomy* **2024**, *14*, 3027. [CrossRef]
4. Wang, X.; Wen, X.; Li, Y.; Du, C.; Zhang, D.; Sun, C.; Chen, B. A Precise Detection Method for Tomato Fruit Ripeness and Picking Points in Complex Environments. *Horticulturae* **2025**, *11*, 585. [CrossRef]
5. Edozie, E.; Shuaibu, A.N.; John, U.K.; Sadiq, B.O. Comprehensive Review of Recent Developments in Visual Object Detection Based on Deep Learning. *Artif. Intell. Rev.* **2025**, *58*, 277. [CrossRef]
6. Cao, Y.; Lu, X.; Zhang, B.; Li, Z.; Zhou, L.; Wu, Q.; Ren, N. A Global–Local Collaborative Image Enhancement Method for Low-Illumination and Occluded Scenes in Robotic Tomato Picking. *Comput. Electron. Agric.* **2026**, *245*, 111537. [CrossRef]
7. Wang, S.; Xiang, J.; Chen, D.; Zhang, C. A Method for Detecting Tomato Maturity Based on Deep Learning. *Appl. Sci.* **2024**, *14*, 11111. [CrossRef]
8. Solimani, F.; Cardellicchio, A.; Dimauro, G.; Petrozza, A.; Summerer, S.; Cellini, F.; Renò, V. Optimizing Tomato Plant Phenotyping Detection: Boosting YOLOv8 Architecture to Tackle Data Complexity. *Comput. Electron. Agric.* **2024**, *218*, 108728. [CrossRef]
9. Zhang, X.; Mamat, S.; Liu, X.; Liu, J.; Liu, R.; Wu, G.; Zhu, P.; Li, H.; Ma, M.; Liu, X. MSRRT-DETR: A High-Precision Apple Detection Method with Strong Cross-Domain Generalization Capability in Complex Orchard Scenes. *PLoS ONE* **2026**, *21*, e0342854. [CrossRef] [PubMed]
10. Sapkota, R.; Karkee, M. Ultralytics YOLO Evolution: An Overview of YOLO26, YOLO11, YOLOv8, and YOLOv5 Object Detectors for Computer Vision and Pattern Recognition. *arXiv* **2025**, arXiv:2510.09653. [CrossRef]
11. Liu, G.; Zhang, Y.; Liu, J.; Liu, D.; Chen, C.; Li, Y.; Zhang, X.; Touko Mbouembe, P.L. An Improved YOLOv7 Model Based on Swin Transformer and Trident Pyramid Networks for Accurate Tomato Detection. *Front. Plant Sci.* **2024**, *15*, 1452821. [CrossRef] [PubMed]
12. Liu, X.; Teng, W.; Yu, H.; Yao, Z.; Wang, C.; Peng, Y.; Han, X.; Liu, J. GAE-YOLO: A Lightweight Multimodal Detection Framework for Tomato Smart Agriculture with Edge Computing. *Front. Plant Sci.* **2025**, *16*, 1712432. [CrossRef] [PubMed]
13. Lin, C.; Jiang, W.; Zhao, W.; Zou, L.; Xue, Z. DPD-YOLO: Dense Pineapple Fruit Target Detection Algorithm in Complex Environments Based on YOLOv8 Combined with Attention Mechanism. *Front. Plant Sci.* **2025**, *16*, 1523552. [CrossRef] [PubMed]
14. Ye, B.; Xue, R.; Xu, H. ASD-YOLO: A Lightweight Network for Coffee Fruit Ripening Detection in Complex Scenarios. *Front. Plant Sci.* **2025**, *16*, 1484784. [CrossRef] [PubMed]
15. Gao, A.; Tian, Z.; Ma, W.; Song, Y.; Ren, L.; Feng, Y.; Qian, J.; Xu, L. Fruits Hidden by Green: An Improved YOLOv8n for Detection of Young Citrus in Lush Citrus Trees. *Front. Plant Sci.* **2024**, *15*, 1375118. [CrossRef] [PubMed]
16. Gao, J.; Zhang, J.; Zhang, F.; Gao, J. LACTA: A Lightweight and Accurate Algorithm for Cherry Tomato Detection in Unstructured Environments. *Expert Syst. Appl.* **2024**, *238*, 122073. [CrossRef]
17. Sunkara, R.; Luo, T. No More Strided Convolutions or Pooling: A New CNN Building Block for Low-Resolution Images and Small Objects. In *Proceedings of the Machine Learning and Knowledge Discovery in Databases*; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2023; Volume 13715, pp. 443–459. [CrossRef]
18. Zhang, H.; Zhang, S. Focaler-IoU: More Focused Intersection over Union Loss. *arXiv* **2024**, arXiv:2401.10525. [CrossRef]
19. Gerz, F.; Jelali, M. Towards Reliable Data Augmentation in Machine Learning: Practices to Prevent Data Leakage. *Smart Agric. Technol.* **2026**, *15*, 102462. [CrossRef]
20. Zheng, S.; Jia, X.; He, M.; Zheng, Z.; Lin, T.; Weng, W. Tomato Recognition Method Based on the YOLOv8-Tomato Model in Complex Greenhouse Environments. *Agronomy* **2024**, *14*, 1764. [CrossRef]
21. Yang, G.; Wang, J.; Nie, Z.; Yang, H.; Yu, S. A Lightweight YOLOv8 Tomato Detection Algorithm Combining Feature Enhancement and Attention. *Agronomy* **2023**, *13*, 1824. [CrossRef]
22. Feng, Q.; Xu, X.; Wang, Z. Deep Learning-Based Small Object Detection: A Survey. *Math. Biosci. Eng.* **2023**, *20*, 6551–6590. [CrossRef] [PubMed]
23. Ma, B.; Hua, Z.; Wen, Y.; Deng, H.; Zhao, Y.; Pu, L.; Song, H. Using an Improved Lightweight YOLOv8 Model for Real-Time Detection of Multi-Stage Apple Fruit in Complex Orchard Environments. *Artif. Intell. Agric.* **2024**, *11*, 70–82. [CrossRef]
24. Liu, S.; Qi, L.; Qin, H.; Shi, J.; Jia, J. Path Aggregation Network for Instance Segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2018; pp. 8759–8768. [CrossRef]
25. Tan, M.; Pang, R.; Le, Q.V. EfficientDet: Scalable and Efficient Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2020; pp. 10781–10790. [CrossRef]

26. Yang, L.; Zhang, R.Y.; Li, L.; Xie, X. SimAM: A Simple, Parameter-Free Attention Module for Convolutional Neural Networks. In *Proceedings of the 38th International Conference on Machine Learning*; PMLR (Proceedings of Machine Learning Research): Cambridge, MA, USA, 2021; pp. 11863–11874.
27. Horn, R.A.; Johnson, C.R. *Matrix Analysis*, 2nd ed.; Cambridge University Press: Cambridge, UK, 2012. [CrossRef]
28. Zheng, Z.; Wang, P.; Liu, W.; Li, J.; Ye, R.; Ren, D. Distance-IoU Loss: Faster and Better Learning for Bounding Box Regression. In *Proceedings of the AAAI Conference on Artificial Intelligence*; AAAI Press: Palo Alto, CA, USA, 2020; Volume 34, pp. 12993–13000. [CrossRef]
29. Ma, S.; Xu, Y. MPDIoU: A Loss for Efficient and Accurate Bounding Box Regression. *arXiv* **2023**, arXiv:2307.07662. [CrossRef]
30. Hu, J.; Shen, L.; Sun, G. Squeeze-and-Excitation Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2018; pp. 7132–7141. [CrossRef]
31. Woo, S.; Park, J.; Lee, J.Y.; Kweon, I.S. CBAM: Convolutional Block Attention Module. In *Proceedings of the Computer Vision—ECCV 2018*; Springer: Cham, Switzerland, 2018; pp. 3–19. [CrossRef]
32. Wang, Q.; Wu, B.; Zhu, P.; Li, P.; Zuo, W.; Hu, Q. ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2020; pp. 11534–11542. [CrossRef]
33. Hou, Q.; Zhou, D.; Feng, J. Coordinate Attention for Efficient Mobile Network Design. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2021; pp. 13713–13722. [CrossRef]
34. Ouyang, D.; He, S.; Zhang, G.; Luo, M.; Guo, H.; Zhan, J.; Huang, Z. Efficient Multi-Scale Attention Module with Cross-Spatial Learning. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*; IEEE: Piscataway, NJ, USA, 2023; pp. 1–5. [CrossRef]
35. Zhang, Y.F.; Ren, W.; Zhang, Z.; Jia, Z.; Wang, L.; Tan, T. Focal and Efficient IOU Loss for Accurate Bounding Box Regression. *Neurocomputing* **2022**, *506*, 146–157. [CrossRef]
36. Gevorgyan, Z. SIoU Loss: More Powerful Learning for Bounding Box Regression. *arXiv* **2022**, arXiv:2205.12740. [CrossRef]
37. Zhang, H.; Zhang, S. Shape-IoU: More Accurate Metric Considering Bounding Box Shape and Scale. *arXiv* **2023**, arXiv:2312.17663. [CrossRef]
38. Zhang, H.; Xu, C.; Zhang, S. Inner-IoU: More Effective Intersection over Union Loss with Auxiliary Bounding Box. *arXiv* **2023**, arXiv:2311.02877. [CrossRef]
39. Tong, Z.; Chen, Y.; Xu, Z.; Yu, R. Wise-IoU: Bounding Box Regression Loss with Dynamic Focusing Mechanism. *arXiv* **2023**, arXiv:2301.10051. [CrossRef]
40. Ren, S.; He, K.; Girshick, R.; Sun, J. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. *Adv. Neural Inf. Process. Syst.* **2015**, *28*, 91–99.
41. Zhao, Y.; Lv, W.; Xu, S.; Wei, J.; Wang, G.; Dang, Q.; Liu, Y.; Chen, J. DETRs Beat YOLOs on Real-Time Object Detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*; IEEE: Piscataway, NJ, USA, 2024; pp. 16965–16974. [CrossRef]
42. Chen, J.; Fu, H.; Lin, C.; Liu, X.; Wang, L.; Lin, Y. YOLOPears: A Novel Benchmark of YOLO Object Detectors for Multi-Class Pear Surface Defect Detection in Quality Grading Systems. *Front. Plant Sci.* **2025**, *16*, 1483824. [CrossRef] [PubMed]
43. Muhammad, M.B.; Yeasin, M. Eigen-CAM: Class Activation Map Using Principal Components. In *Proceedings of the International Joint Conference on Neural Networks*; IEEE: Piscataway, NJ, USA, 2020; pp. 1–7. [CrossRef]
44. Ci, J.; Wang, X.; Rapado-Rincón, D.; Burusa, A.K.; Kootstra, G. 3D Pose Estimation of Tomato Peduncle Nodes Using Deep Keypoint Detection and Point Cloud. *Biosyst. Eng.* **2024**, *243*, 57–69. [CrossRef]
45. Kang, H.; Zhou, H.; Wang, X.; Chen, C. Real-Time Fruit Recognition and Grasping Estimation for Robotic Apple Harvesting. *Sensors* **2020**, *20*, 5670. [CrossRef] [PubMed]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.